

Inter-UAV Collision Avoidance using Deep-Q-Learning in Flocking Environment

Gunasekaran Raja

Department of Computer Technology
Anna University
Chennai, India
dr.r.gunasekaran@ieee.org

Sudha Anbalagan

Department of Computer Science and Engineering
SRM Institute of Science and Technology
Chennai, India
sudhaa@srmist.edu.in

Vikraman Sathiya Narayanan

Department of Computer Technology
Anna University
Chennai, India
vikikkdi@gmail.com

Srinivas Jayaram

Department of Computer Technology
Anna University
Chennai, India
srinivasbenten1998@gmail.com

Aishwarya Ganapathisubramanian

Department of Computer Technology
Anna University
Chennai, India
aishwarya97.mit@gmail.com

Abstract—In the past twenty years, Unmanned Aerial Vehicles (UAVs) have demonstrated their ability in supporting both military and civilian applications. It has proved itself useful in tasks which are dangerous or too costly with usual methods. Certain applications also contains tasks which can be processed faster by using multiple UAVs that work together and perform smaller tasks which can be further combined to get the intended result. However, to achieve this, we need autonomous coordination among the multi UAVs. Multiple UAVs flying as a swarm, incorporates a pattern or a shape that has to be maintained through out the flight. But there can be cases where the pattern or shape of swarm has to be changed. Under such circumstances, collision has to be avoided between the UAVs as they travel to their new position forming a new shape. We have proposed an algorithm which helps in finding optimal goal positions for the UAVs to flock and to find the trajectory which supports collision free movement to its assigned position. We also ensure that the total distance travelled by individual UAV is minimized. The optimal goal position assignment is done using Hungarian Algorithm. Deep Q Learning method is used to find the optimal flight parameters for a collision free trajectory for the UAVs to reach its goal. Results from simulation show that the algorithm is sufficiently fast for practical applications as optimal assignments and flight parameters were computed for 50 UAVs in less than 1s.

Index Terms—Unmanned Aerial Vehicle, Deep Q Learning, Flocking, Swarming, Hungarian Algorithm, Obstacle Avoidance, Collision Avoidance

I. INTRODUCTION

Obstacle detection and avoidance in UAVs has been an emerging topic of research over the decade. Several algorithms have been introduced for detection and avoidance of obstacles. A wide array of sensors have been employed for detecting obstacles like the kinetic depth sensor [1], Infrared (IR)[2]. Input from camera is also used for detecting obstacles [3][4] where obstacle detection is done by using Convolutional Neural Network (CNN) which extracts relevant features from the images and aids in object detection. Vision based techniques also include image processing, background subtraction for

obstacle detection. Reinforcement Learning (RL) techniques has been used for obstacle avoidance in UAVs where the model learns optimal strategy to avoid obstacle while maintaining the flight parameters.

In these obstacle detection and avoidance problem the position of the obstacle is not known and so, algorithms are developed to identify the relative position of the obstacle with respect to UAV based on camera position and relative angle of the obstacle. But our work addresses a much specific case of obstacle avoidance [5]. In flocking environment, multiple UAVs can form a shape. In a leader-follower topology, the master UAV moves along the predefined trajectory and the follower UAVs move along with the master, maintaining the relative position with respect to the master UAV. When UAVs are instructed to change the shape of the pattern, all the UAVs try to reach its calculated goal position. During this course of movement, there is a possibility that there may be collisions even within the UAVs. In our research work, we specifically address this issue of inter-UAV collision avoidance. The advantage of such collision avoidance over the general obstacle detection and avoidance is that, we know the exact location of each of the UAV from the Ground Control Station (GCS). UAVs communicate its movement with the GCS and it acts as a controller from which humans can interact with the UAVs. The information from the GCS can be exploited to our use [6].

Using the information about the positions of the UAVs in the swarm environment from the GCS, the optimal assignment is calculated for the goal shape formation using Hungarian algorithm. We have trained a Deep-Q-Learning based RL model to adjust the goal formation positions such that the total distance travelled by the UAV is minimized. After the final goal positions are fixed, our proposed neural network generates obstacle-less trajectories which leads to the new goal positions.

The above mentioned algorithm is independent of the num-

ber of UAVs being used. We also present the algorithm for computing the goal formation position for few shapes like square, circle, V-shape, a matrix to which the number of UAV is given as a input. The model developed is not restricted to any UAV type. As a result, the dynamics of the vehicle needn't be considered for training process. It can be used for all kinds of robots including fixed wing UAVs.

In our work, we have identified the need for collision avoidance between multiple UAVs in a swarming environment and designed Deep-Q-Learning methodologies for planning the path for UAVs to its assigned goal positions that were computed using the Hungarian Algorithm. The existing methods for path planning in swarm environment, obstacle detection and avoidance has been carefully studied and has been presented in literature survey (Section II). Section III provides a basic introduction on reinforcement learning. Section IV gives the proposed work followed by the necessary details about the implementation of the algorithm (Section V). Simulation results of the algorithm is shown in Section VI. Conclusion for our research work and future work are discussed in detail in Section VII.

II. LITERATURE SURVEY

In [7], the author works on parallel optimization of goal assignments and shape formations for several robots. Here the author derives algorithms to find out the optimal assignments and the parameters for goal formation for a group of robots from their initial formation to a changing goal formation (where the shape of the goal pattern is given, and parameters like scale and location are to be optimized). Authors considers n robots are in shape of identical spheres. Sum of squared travel distances was incorporated as the objective function to be minimized, which also makes sure that generated paths are collision free trajectories. This assignment with variable goal formation problem can be converted to a Linear Sum Assignment Problem (LSAP) with pseudo costs. The transformed problem can be solved using the Hungarian algorithm in $\mathcal{O}(n^3)$ time.

Forest has been an important environment where autonomous exploration and navigation is desirable as these places do not have access to GPS for localization of the drone. In [8], authors address the formation control for a team of UAVs in which the UAVs follow a specified swarm trajectories while safely changing the shape of the swarm when instructed to do so without any collisions. Each of the UAV individually plans its trajectory based on the local information of the neighbours it has which includes the trajectory planned by its neighbor and its states [9][10]. In [3], the authors proposed a CNN based obstacle avoidance system. The CNN was trained with a limited dataset of forest environment. Very deep convolutional neural network [11][12] methods have been used for image classification for larger dataset like ImageNet and small dataset like CIFAR-10. Transfer learning methods can be used for the classification by using already pretrained model like VGG-16, AlexNet, Inception etc. This method has shown to reduce the error rate by significant amounts. In [13], the

author discusses flocking of UAVs in stochastic environment as a model free RL problem. Q Learning method of RL that was introduced by Peng. In [2], a multi scale CNN architecture for IR foreground object detection. With informations from multiple scale, the architecture extracted a good number of features with fine-grain details. Micro Aerial Vehicles (MAV) are very suitable for flying in the indoor environment but autonomous navigation in indoor can be difficult as MAVs have strict hardware limitations. In [14], author proposes an algorithm known as Edge-FS for the determination of velocity and depth using the input from the camera. This method allows the MAV to control its velocity, other flight attitudes and also avoid obstacles. In [1], a computer vision-based algorithm is proposed using kinetic depth sensor for autonomous navigation of vehicles.

AlexNet [15] was used for getting the informations/features from the images. The network also predicted the direction that the UAV has to take in-order to avoid the tree during its course of flight [16]. Background subtraction has been used for a long time for detecting objects in an image using IR light. Because of certain characteristics of IR light, a few algorithms can be used for IR foreground object detection. CNN has seen a tremendous growth in the recent years. CNN has been used for classification of images for over a decade. CNNs architecture can be modified by removing the dense layer at the end and using a deconvolution layer to produce segmented images. This architecture, also called as Fully Convolutional network (FCN) [4] produces pixel-wise classification of the input image [17][18]. Using the pixel wise, classification, we can detect obstacles and plan a path that would avoid those obstacles.[19] focuses mainly on vision based obstacles avoidance of UAVs. Detection of the obstacle in the environment and controlling the UAV to avoid the detected obstacles are the two key components of sense and avoid system. A bio inspired human vision algorithm known as saliency[20] is used to automatically detect objects from input images. Furthermore, the author also proposes an yaw angle calculator algorithm which relies on the geometrical relationship between the UAV and its camera.

In [21], authors proposes an online approach for detection of Small UAVs and also to estimate their relative position and velocities in 3D environment using a single moving camera. The author uses a AdaBoost approach, a multi staged framework that incorporates fast object detection algorithm with an online vision based tracking algorithm.

Optimal goal formation uses distributed auction assignment, that can vary the number of target positions in assignment and sequential convex programming is used for generation of collision free trajectories in [22]. The model predictive control formulation uses current state measurements to solve the optimal assignment and form a trajectory. Once the given shape is decided and updated to the swarm, the algorithm determines where each robot must reach the position in a collision free manner. In [23], author focuses on the problem of assignment of goal position for variable robot formation. Initially, cost is calculated for the travel of each robot to each

of the goal position available using computational geometry techniques. Then a minimum completion time assignment for an equivalence class is computed by formulating it as a Linear Bottleneck Assignment Problem (LBAP).

In a leader-follower topology, the follower UAVs use variable learning rate to learn a control policy that helps them in flocking [24]. This problem is structured as a Markov decision process, where the agents are modeled as UAVs that experience stochasticity because of various issues like weather, which causes weight and balance issues. Q Learning [24] is a type of RL algorithm in which the agent forms a table during the process of training and uses the table to find the optimal action that can be taken in the given state at runtime.

III. REINFORCEMENT LEARNING

RL is a type of machine learning algorithm which is semi-supervised. It involves taking a decision on the suitable action to be performed on the environment for a given set of input such that the reward attained by the agent is maximized. It is used to find the best possible path or behaviour that the agent should take in a given environment. Difference between RL and supervised learning is that in RL, the agent doesn't know about the correct action to be performed for a given input. Instead it tries to learn the optimal action based on the reward it gets. In supervised learning the training data has the correct action with which the model is initially trained. Because of the absence of training dataset, the agent will learn from its experience.

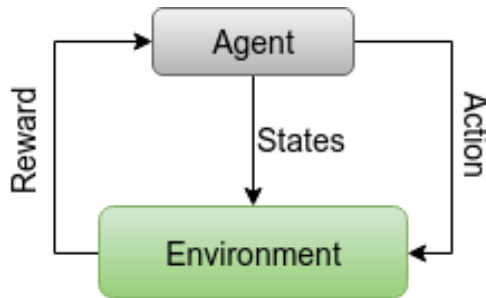


Fig. 1. Interaction between an agent and environment

Fig. 1 shows the basic interactions between the agent and the environment. Environment is the place where the agent acts on. In our case, it's an obstacle-less swarm of UAVs. The agent perceives input from the environment and acts on it using its experience and the current state in which it is present. The agent uses a reward function to compute the reward for an action and it finally chooses the best action to carry out based on the available information.

A. Q-Learning

Q-Learning is a model-free reinforcement algorithm. Its goal is to learn a policy that helps the agent in deciding the actions that have to be taken under different circumstances. The agent's main motive is to maximize the value for the reward function for the given set of inputs. It achieves this by finding the sum

of maximum reward that it can attain from the future states when it performs an action to move to the current state. Thus influencing the action performed now on the environment from the possible future states. This potential reward is a weighted sum of the expected values of the rewards of all future steps starting from the current state. These rewards are stored in a simple look-up table known as the Q-Table. This table guides the agent to the best action at each state. The calculation for the Q table ($Q^\pi(s_t, a_t)$) is as per equation (1) where t is the time, $s_t \in S$ denotes the state at time t , $a_t \in A$ denotes the action at time t , γ is the learning rate ($0 < \gamma \leq 1$), R_t is the reward at time t .

$$Q^\pi(s_t, a_t) = E[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | s_t, a_t] \quad (1)$$

One thing to think about is that, the state-action space will be very big for most of the problems. So, having a look-up table for finding an action given a state can be difficult in certain conditions.

B. Deep Q-Learning

The difference between Q-Learning and Deep Q-Learning is that, we use a deep neural network architecture to approximate the Q value for each action based on the state. Deep Q Network (DQN) introduces experience replay and target network to slow down the changes so we can learn Q gradually. Experience replay stores the few state-action-reward in a replay buffer that the agent has encountered previously. We train Q with batches of random samples from the replay buffer. Therefore, the training samples are randomized. This behaves closer to the supervised learning in Deep Learning.

In DQN for inter-UAV collision avoidance in a swarm environment, the model is trained where each of the current positions of the UAV can be taken as the state for RL. Finally, the RL agent tries to learn the optimal position to place the goal formation so that the total distance travelled by individual UAV is reduced.

IV. PROPOSED WORK

The basic workflow for the inter-UAV collision avoidance in a swarm environment using DQN is shown in Fig. 2.

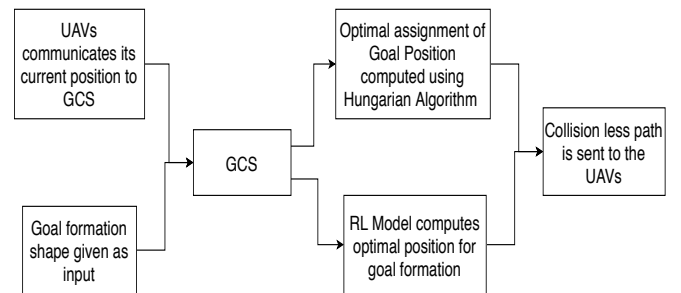


Fig. 2. Work flow of inter-UAV collision avoidance in swarm using DQN

Initially, the UAVs communicate their current location to the GCS. The GCS is provided with the shape and the relative distance between the UAVs for goal formation. The GCS

uses this information for computing the optimal assignment of UAVs to its goal formation using the Hungarian Algorithm. Then, the trained RL model is used to move the goal formation position such that the distance travelled by every individual UAV is minimized.

The RL model is trained with a neural network whose input layer consists of 21 neurons as input. The values of 21 neurons include x, y, z, roll, pitch, yaw angle of the UAV and their first, second order derivatives. The first order derivative gives the velocity along the axis and second order derivative gives the acceleration along the axis. The first hidden layer consists of 30 neurons and the second hidden layer has 20 neurons. The final layer (output layer) consists of 3 neurons which provides the x, y, z which corresponds to the new goal position for the UAV.

Once the Hungarian algorithm computes the optimal goal position assignment to the UAVs, the above neural network is used to change the position of the goal formation relative to the current UAVs position as well as the flight attitude parameters (roll, pitch, yaw) which helps in reducing the distance travelled by each UAV and also provides a collision free path for the UAVs to reach its assigned goal position.

V. IMPLEMENTATION DETAILS

A. Swarm of Drones

While UAVs are moving as a swarm with leader follower topology, there may be instances where we may have to change the formation pattern of the UAVs. For example, suppose 10 UAVs are flying in a leader follower topology in a triangle shape and we have to change the formation to diamond, there is a need for obstacle avoidance among the UAVs when changing the formation. The difference between general obstacle avoidance and this problem stated above is the, in general case we do not have the location of the obstacle with us but this is not the case with swarming. When moving in a leader follower scenario, each UAV communicates its current location to the ground control station. The GCS can then act on obstacle avoidance with the added information of each UAVs location. This makes it easy for a reinforcement learning model to learn the optimal strategy to avoid obstacle in such instances.

Lets consider N UAVs standing in a matrix formation of an initial dimension. Positions of each of the UAV can be taken from the GCS. Let the position of a UAV be a vector [x, y, z]. The matrix P has positions of all the UAVs and its dimension is Nx3 where N is the number of UAVs and 3 corresponds to the x, y and the z axis. Based on the implementation, the position of each vehicle can be taken either as the latitude, longitude, height tuple that would give the absolute position of each vehicle or x, y, z relative to the reference point which is our GCS at [0, 0, 0]. The matrix P would give the initial formation of the UAVs.

Lets consider a matrix S of Nx3 dimension that gives the goal shape that has to be formed by the vehicles (for example a diamond) where N is the number of UAVs and 3 corresponds to the axes that would give the points for the

required shapes. To have an obstacle avoidance system in such a unique environment, we have two goals.

Goal 1: Assign each of the N UAV to each of the N goal position i.e., find an optimal allocation of UAV where UAV[i] is assigned to GOALPOSITION[j] such that the cost for the travel is reduced. Here the cost function we use is distance equation as given by equation 2.

$$J(w) = \sum_{i=0}^n \sum_{j=0}^n \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (2)$$

Goal 2: Move the Goal formation and scale it in such a way that the cost for the UAV is reduced.

To accompany Goal 1, we compute a cost matrix which is NxN matrix that has the distance between the UAV[i] and GOALPOSITION[j] in COST[i][j]. Then the assignment problem can be reduced to a Linear Sum Assignment problem where there are N UAV and N goal position and each UAV has to be allocated to a Goal position such that the cost is reduced. For this purpose, we use Hungarian Algorithm. The output of Hungarian algorithm consists of an matrix Nx2 where ith entry of RESULT states that RESULT[i][0] UAV is allocated to RESULT[i][1] goal position.

To accompany Goal 2, we use gradient descent and Deep reinforcement learning where the model learns to place the goal positions optimally such that there are no obstacles when the UAVs move to their respective goal positions. This is a model-less reinforcement learning problem where the q value is optimized with the help of a fully connected neural network. The loss of the neural network is reduced with the help of gradient descent where the loss function is given by equation 2. The architecture of neural network that is used for changing the position of goal formation for having an obstacle-less path is used.

The output values (x, y, z) given by the neurons are used to move the Goal formation position such that the UAV assigned to each of the goal positions move to its corresponding destination without any obstacles in its path during the course of its motion. This also ensures that the total distance travelled by the UAVs is minimized.

After the UAVs are assigned to its goal positions and the new goal locations are formed, we can send command to the UAVs to move to its corresponding destination thus forming the required shape.

B. Reward Function

Reward functions are important for any RL problem as the agent tries to learn the optimal strategy using this. The reward function for our problem statement uses the current location of each of the UAVs to navigate in the environment without any collisions. The reward is calculated such that, if there are any collision, the agent is given with a huge penalty (a negative score). If there is no collision but the agent failed to reach its assigned goal formation position, then the agent is given with a low penalty as the main objective of the RL agent is to form the trajectories of UAVs without collision. If the agent's action avoids all the collisions among UAVs and if all

the UAVs reaches its assigned goal formation position, then a reward (a positive score) is given to the agent.

C. Hungarian Algorithm

The Hungarian method is a combinatorial optimization algorithm that solves the assignment problem in polynomial time. In a system with N workers and N jobs where $COST[i][j]$ gives the cost for doing work j by the worker i, Hungarian Algorithm is used to assign each worker with a job such that the total cost of doing the work is minimized. In inter-UAV collision avoidance, this is used to assign each of the UAV with the goal formation position such that the total distance travelled by the UAV is reduced. The pseudo-code for calculating the optimal assignment using Hungarian Algorithm is given in algorithm 1.

Algorithm 1 Optimal Assignment of Goal Formation Position

```

1:  $k \leftarrow \{n * n \text{ matrix with zeros}\}$ 
2:  $i, j \leftarrow \{0\}$ 
3: while  $i < n$  do
4:   while  $j < n$  do
5:      $k[i][j] \leftarrow -P[i]^T * S[j]$ 
6:     increment j
7:   end while
8:   increment i
9: end while
10:  $X^*, K^* \leftarrow \text{Hungarian} - \text{Algorithm}(k)$ 
11: return  $[X^*, K^*]$ 

```

D. Shape Formations

Simulation has been done for shapes such as star, square, V, circle, matrix where the input given is N, denoting the number of UAVs and the function computes the N points such that the points are along the perimeter of the shape and are equidistant.

Algorithm 2 provides the necessary steps to generate square for N UAVs. The algorithm fixes the coordinates for the four corners initially and then divides each of the edges into two by introducing one point between them until all N points are generated. The 11th line of the algorithm plays a vital role in forming the square. It ensures that the points generated are equidistant and uniformly distributed.

Algorithm 3 provides the necessary steps to generate circle for N UAVs. The algorithm gets an input from user for the radius of the circle. Then it computes cos and sin values for various points to generate the necessary coordinates for N UAVs along the circumference of the circle.

Shape like matrix can be computed by finding the optimal factors for the number of UAVs N such that the product of those two factors is equal to N. One of the factor gives the number of UAVs to be placed along the row and the other gives the number of UAVs to be placed along the column. Then the coordinates for placing each of the drone can be computed by having a uniform difference between each UAV along the row and column. The two factors of N which multiplies to give

Algorithm 2 Generating Coordinates for Square

```

1:  $sq \leftarrow \emptyset$ 
2:  $N \leftarrow \text{number of UAVs}$ 
3:  $tx, ty \leftarrow 100, 100$ 
4:  $bx, by \leftarrow 500, 500$ 
5:  $sq.append([tx, ty], [bx, ty], [bx, by], [tx, by])$ 
6:  $N \leftarrow N - 4$ 
7:  $i \leftarrow 0$ 
8: while  $n \neq 0$  do
9:    $x \leftarrow sq[i][0] + sq[i + 1 \% len(sq)][0] / 2$ 
10:   $y \leftarrow sq[i][1] + sq[i + 1 \% len(sq)][1] / 2$ 
11:   $sq \leftarrow sq[:i + 1] + [[x, y]] + sq[i + 1 :]$ 
12:   $i \leftarrow i + 2 \% len(sq)$ 
13:   $N \leftarrow N - 1$ 
14: end while
15: return  $sq$ 

```

Algorithm 3 Generating Coordinates for Circle

```

1:  $circle \leftarrow \emptyset$ 
2:  $N \leftarrow \text{number of UAVs}$ 
3:  $r \leftarrow \text{radius of the circle}$ 
4:  $i \leftarrow 0$ 
5: while  $n \neq 0$  do
6:    $x \leftarrow \cos(2 * \pi / n * i) * r$ 
7:    $y \leftarrow \sin(2 * \pi / n * i) * r$ 
8:    $circle \leftarrow circle + [x, y]$ 
9:    $N \leftarrow N - 1$ 
10:   $i \leftarrow i + 1$ 
11: end while
12: return  $circle$ 

```

N can be computed by finding the prime factors of N and dividing it into two sets of approximately uniform product.

Line shape can be formed by having a start point and adding an uniform value along the x and y axis so that each UAV is separated from the other with a uniform distance.

VI. RESULTS

We have implemented the obstacle avoidance in swarm environment where we know the coordinates, speed and the attitude of all other UAVs. Simulation has been done with 10 UAVs simultaneously moving towards its goal position without colliding with other UAVs while also changing swarm pattern from one formation to another. The algorithm for assigning the UAV to its corresponding goal position using Hungarian Algorithm took approximately 5ms when the simulation was made with 10 drones. The algorithm for finding the optimal goal position using the deep reinforcement learning took 12ms when the simulation was made using 10 drones. The graph for the performance of Hungarian algorithm based on the number of drones has been given below in the Fig 3.

The performance of the algorithm for finding the optimal goal position using the deep q-learning when formation is

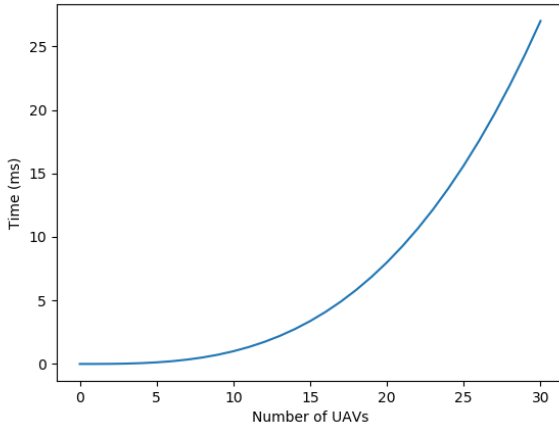


Fig. 3. Performance of Hungarian Algorithm

changed from 2x5 matrix to Star shape in 10 drone environments is given below in the Fig 4.

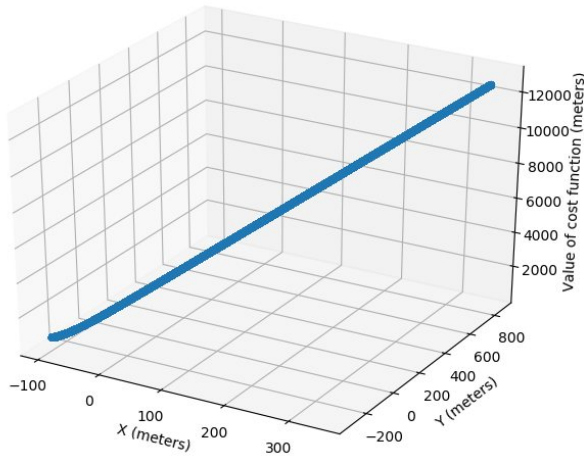


Fig. 4. Cost function values for initial position to star shape

Fig 5 shows the different types of arrangement for the swarm behaviour. Simulation results show that, the DQN model trained for shape transformation from matrix (Fig 5[b]) to circle (Fig 5[a]) can be used for transformation between any shapes. Thus the model trained is independent of the goal formation shape and can be used for dynamic shape formations as well.

Fig 6 shows the trajectory for 10 UAVs in a swarming environment having an initial shape of Matrix (2x5) with a goal formation shape of a Star. The trajectories, generated by the DQN model, shows that the trajectories are collision free. Similarly the Fig 7 shows the trajectory for 10 UAVs having an initial shape of Star with the goal formation shape of a Circle.

In Fig 6 and Fig 7, the white circle denotes the initial shapes and the colored circles defines the goal formation position.

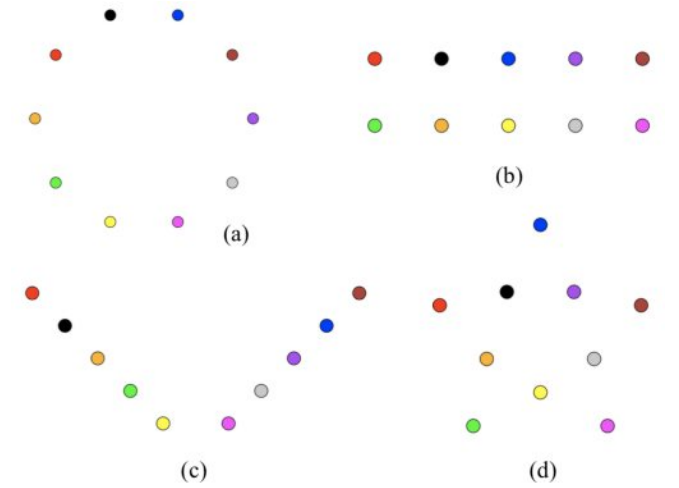


Fig. 5. Swarm using 10 UAVs. (a) UAVs arranged in Circle shape (b) UAVs arranged in Matrix shape (c) UAVs arranged in V shape (d) UAVs arranged in Star shape

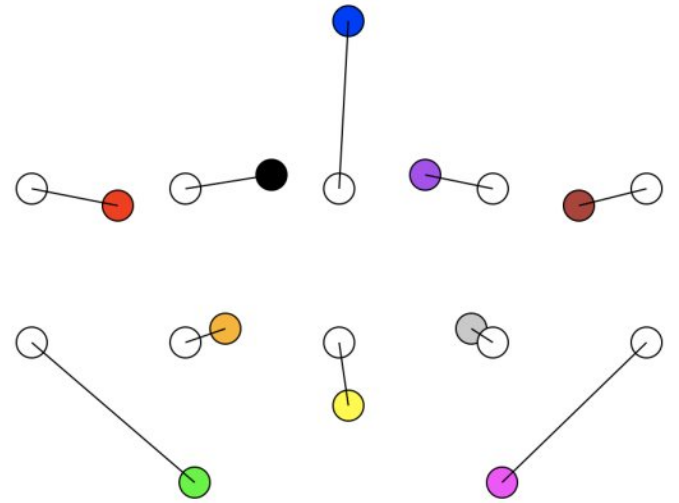


Fig. 6. Trajectory for UAVs moving from Matrix shape to Star shape

The straight lines defines the trajectories of the UAVs when moving from initial position to goal formation position.

VII. CONCLUSION

Deep Q-learning and Hungarian algorithm has been used for implementing obstacle avoidance in multi-UAV swarm environment, where each UAV has the knowledge about the other UAVs coordinates and attitude. The proposed implementation scales well up-to 100 drones after which the time taken for forming the goal position takes a huge amount of time. When the time taken for forming the goal position is huge, it can be a set-back in an online system where decisions has to be made spontaneously. From the computed results, we also proved that the distance travelled by the individual UAV gets reduced considerably. Future work includes optimizing the algorithm proposed in this paper to scale for more than 200 drones so that the path planning in an online setting takes very little

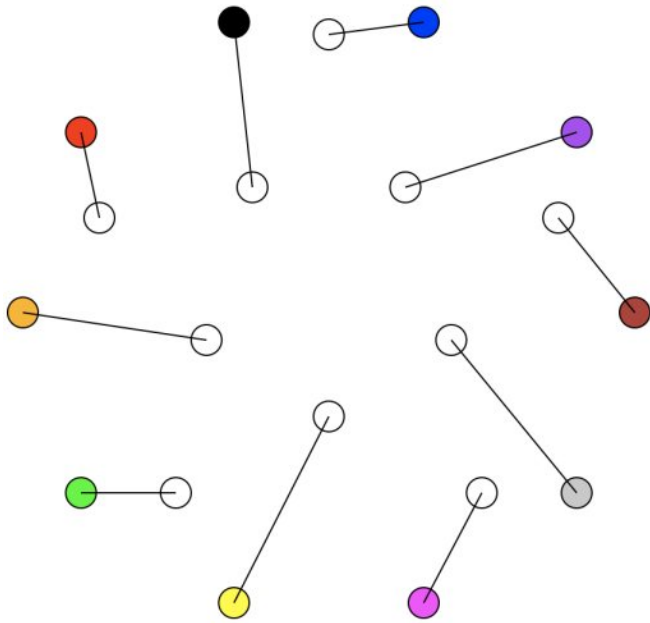


Fig. 7. Trajectory for UAVs moving from Star shape to Circle shape

amount of time as close as few milliseconds.

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REFERENCES

- [1] N. Eric and J. Jang, "Kinect depth sensor for computer vision applications in autonomous vehicles", *IEEE Conferences*, pp. 531-535, 2017.
- [2] D. Zeng and M. Zhu, "Multiscale Fully Convolutional Network for Foreground Object Detection in Infrared Videos", *IEEE Journals and Magazines*, vol. 15, no.4, pp. 617-621, 2018.
- [3] S. Dionisio-Ortega, L. O. Rojas-Perez, J. Martinez-Carranza and I. Cruz-Vega, "A deep learning approach towards autonomous flight in forest environments", *IEEE Conference*, pp. 139-144, 2018.
- [4] E. Shelhamer, J. Long and T. Darrell, "Fully Convolutional Networks for Semantic Segmentation", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 4, pp. 640-651, 2017.
- [5] U. Muller, J. Ben, E. Cosatto, B. Flepp, Y. L. Cun, "Off-road obstacle avoidance through end-to-end learning", *Advances in neural information processing systems*, pp. 739-746, 2006.
- [6] Gunasekaran Raja, Aishwarya Ganapathisubramanian, Gokul Anand and Gowshika, "Intrusion Detector for Blockchain based IoT Networks," in *Proc. 10th IEEE International Conference on Advanced Computing (ICoAC)*, Chennai, Dec. 2018, pp. 328332.
- [7] S. Agarwal and S. Akella, "Simultaneous Optimization of Assignments and Goal Formations for Multiple Robots", *IEEE International Conference on Robotics and Automation (ICRA)*, Brisbane, QLD, pp. 6708-6715, 2018.
- [8] M. Turpin, N. Michael, and V. Kumar, "Decentralized formation control with variable shapes for aerial robots", *IEEE International Conference on Robotics and Automation*, pp. 2330, 2012.
- [9] R. Wolf and J. C. Platt, "Postal address block location using a convolutional locator network", *Advances in Neural Information Processing Systems*, pp. 745745, 1994.
- [10] M. V. Srinivasan, M. Lehrer, W. H. Kirchner, and S.W.Zhang, "Range perception through apparent image speed in freely flying honeybees", *Visual Neuroscience*, vol. 6, pp. 519-535, 1991.
- [11] S. Liu and W. Deng, "Very deep convolutional neural network-based image classification using small training sample size", *3rd IAPR Asian Conference on Pattern Recognition (ACPR)*, Kuala Lumpur, pp. 730-734, 2015.
- [12] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition". *CoRR*, abs/1409.1556, 2014.
- [13] S. Hung and S. N. Givigi, "A Q-Learning Approach to Flocking With UAVs in a Stochastic Environment", *IEEE Transactions on Cybernetics*, vol. 47, no. 1, pp. 186-197, 2017.
- [14] K. McGuire, G. de Croon, C. De Wagter, K. Tuyls and H. Kappen, "Efficient Optical Flow and Stereo Vision for Velocity Estimation and Obstacle Avoidance on an Autonomous Pocket Drone", *IEEE Journals and Magazines*, vol. 2, pp. 1070-1076, 2017.
- [15] Krizhevsky, A., Sutskever, I., and Hinton, G. E, "ImageNet classification with deep convolutional neural networks", *NIPS*, pp. 11061114, 2012.
- [16] A. Krizhevsky, G. Hinton, "Learning multiple layers of features from tiny images", *Computer Science Department University of Toronto Tech. Republic*, vol. 1, no. 4, pp. 7, 2009.
- [17] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke and Andrew Rabinovich, "Going deeper with convolutions", *IEEE Conferences*, pp. 1-9, 2015.
- [18] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks", *Computer Vision-ECCV Springer*, pp. 818-833, 2014.
- [19] Z. Ma, T. Hu, L. Shen, W. Kong and B. Zhao, "A detection and relative direction estimation method for UAV in sense-and-avoid", *IEEE International Conference on Information and Automation*, Lijiang, pp.2677-2682, 2015.
- [20] J. Lei et al., "A Universal Framework for Salient Object Detection", *IEEE Transactions on Multimedia*, vol. 18, no. 9, pp. 1783-1795, 2016.
- [21] K. R. Sapkota et al., "Vision-based Unmanned Aerial Vehicle detection and tracking for sense and avoid systems", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Daejeon, 2016, pp. 1556-1561.
- [22] D. Morgan, G. P. Subramanian, S.-J. Chung, and F. Y. Hadaegh, "Swarm assignment and trajectory optimization using variable-swarm, distributed auction assignment and sequential convex programming", *The International Journal of Robotics Research*, vol. 35, no. 10, pp. 12611285, 2016.
- [23] S. Akella, "Assignment algorithms for variable robot formations", *12th International Workshop on the Algorithmic Foundations of Robotics*, 2016.
- [24] Peng, J., Williams and R.J. Mach Learn, "Incremental multi-step Q-learning", *IEEE Conference*, 1996